

Slip 1

1)Write a Java program to print the sum of elements of the array. Also display array elements in ascending order.

```
import java.util.Scanner;

public class ArraySortSum {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of elements: ");

        int n = sc.nextInt();

        int[] a = new int[n];

        int sum = 0;

        System.out.println("Enter array elements:");

        for (int i = 0; i < n; i++) {

            a[i] = sc.nextInt();

            sum = sum + a[i];

        }

        for (int i = 0; i < n - 1; i++) {

            for (int j = 0; j < n - i - 1; j++) {

                if (a[j] > a[j + 1]) {

                    int temp = a[j];

                    a[j] = a[j + 1];

                    a[j + 1] = temp;

                }

            }

        }

    }

}
```

```

    }
}

System.out.println("Sum of array elements = " + sum);
System.out.println("Array elements in ascending order:");
for (int i = 0; i < n; i++) {
    System.out.print(a[i] + " ");
}
}
}

```

Slip 2

1) Write a Java Program to Display Armstrong Numbers Between range. Accept range from user.

```

import java.util.Scanner;

public class ArmstrongRange {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter starting range: ");

        int start = sc.nextInt();

        System.out.print("Enter ending range: ");

        int end = sc.nextInt();

        System.out.println("Armstrong numbers between " + start + " and " + end +
        ":");
    }
}

```

```

for (int num = start; num <= end; num++) {
    int temp = num;
    int sum = 0;
    int digits = 0;
    while (temp != 0) {
        digits++;
        temp = temp / 10;
    }
    temp = num;
    while (temp != 0) {
        int digit = temp % 10;
        sum = sum + (int) Math.pow(digit, digits);
        temp = temp / 10;
    }
    if (sum == num) {
        System.out.print(num + " ");
    }
}
}

```

Slip 3

1) Write a package for String operation which has two classes Con and Comp. Con class has to concatenate two strings and comp class compares two strings. Also display proper messages on execution.

1. Create Package: StringOperation

Step 1: mkdir -p JavaProject/StringOperation

Step 2: cd JavaProject

Step 3: vim StringOperation/Con.java

```
package StringOperation;

public class Con {
    public void concatenate(String s1, String s2) {
        System.out.println("Concatenated String: " + s1 + s2);
    }
}
```

Step 4: vim StringOperation/Comp.java

```
package StringOperation;

public class Comp {

    public void compare(String s1, String s2) {

        if (s1.equals(s2))

            System.out.println("Both strings are equal.");

        else

            System.out.println("Both strings are not equal.");

    }

}
```

Step 5: vim StringDemo.java

```
import java.util.Scanner;

import StringOperation.Con;
```

```
import StringOperation.Comp;

public class StringDemo {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");

        String s1 = sc.nextLine();

        System.out.print("Enter second string: ");

        String s2 = sc.nextLine();

        Con c = new Con();

        Comp cp = new Comp();

        c.concatenate(s1, s2);

        cp.compare(s1, s2);

    }

}
```

Step 6 : javac StringOperation/Con.java StringOperation/Comp.java
StringDemo.java

OR

javac StringDemo

Step 7 : java StringDemo

Slip 4

1)Write a menu driven program to perform the following operations on multidimensional array ie matrix:

i. Addition


```
int c1 = sc.nextInt();

int[][] a = new int[r1][c1];

int[][] b = new int[r1][c1];

int[][] sum = new int[r1][c1];

System.out.println("Enter first matrix:");

for (int i = 0; i < r1; i++) {

    for (int j = 0; j < c1; j++) {

        a[i][j] = sc.nextInt();

    }

}

System.out.println("Enter second matrix:");

for (int i = 0; i < r1; i++) {

    for (int j = 0; j < c1; j++) {

        b[i][j] = sc.nextInt();

    }

}


for (int i = 0; i < r1; i++) {

    for (int j = 0; j < c1; j++) {

        sum[i][j] = a[i][j] + b[i][j];

    }

}

System.out.println("Addition of matrices:");
```

```
for (int i = 0; i < r1; i++) {  
    for (int j = 0; j < c1; j++) {  
        System.out.print(sum[i][j] + " ");  
    }  
    System.out.println();  
}  
break;
```

case 2:

```
System.out.print("Enter rows of first matrix: ");  
int r2 = sc.nextInt();  
System.out.print("Enter columns of first matrix: ");  
int c2 = sc.nextInt();  
System.out.print("Enter rows of second matrix: ");  
int r3 = sc.nextInt();  
System.out.print("Enter columns of second matrix: ");  
int c3 = sc.nextInt();  
if (c2 != r3) {  
    System.out.println("Matrix multiplication is not possible.");  
    break;  
}  
  
int[][] m1 = new int[r2][c2];  
int[][] m2 = new int[r3][c3];  
int[][] product = new int[r2][c3];
```

```
System.out.println("Enter first matrix:");

for (int i = 0; i < r2; i++) {
    for (int j = 0; j < c2; j++) {
        m1[i][j] = sc.nextInt();
    }
}

System.out.println("Enter second matrix:");

for (int i = 0; i < r3; i++) {
    for (int j = 0; j < c3; j++) {
        m2[i][j] = sc.nextInt();
    }
}

for (int i = 0; i < r2; i++) {
    for (int j = 0; j < c3; j++) {
        for (int k = 0; k < c2; k++) {
            product[i][j] += m1[i][k] * m2[k][j];
        }
    }
}

System.out.println("Multiplication of matrices:");

for (int i = 0; i < r2; i++) {
    for (int j = 0; j < c3; j++) {
```

```

        System.out.print(product[i][j] + " ");
    }

    System.out.println();
}

break;
case 3:

    System.out.print("Enter rows: ");

    int r4 = sc.nextInt();

    System.out.print("Enter columns: ");

    int c4 = sc.nextInt();

    int[][] matrix = new int[r4][c4];


    System.out.println("Enter matrix:");

    for (int i = 0; i < r4; i++) {
        for (int j = 0; j < c4; j++) {
            matrix[i][j] = sc.nextInt();
        }
    }

    System.out.println("Transpose of matrix:");

    for (int j = 0; j < c4; j++) {
        for (int i = 0; i < r4; i++) {
            System.out.print(matrix[i][j] + " ");
        }
    }

```

```

        System.out.println();
    }
    break;
case 4:
    System.out.println("Program terminated.");
    break;
default:
    System.out.println("Invalid choice. Please try again.");
}
} while (choice != 4);
sc.close();
}
}

```

Slip 5

1) Write a program to accept a text file from user and display the contents of a file in reverse order and change its case.

sample File: test.txt

Hello World
Java Programming

Main program

```
import java.io.*;
```

```
import java.util.Scanner;

public class FileReverseCase {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter file name: ");

        String fileName = sc.nextLine();

        try {

            File file = new File(fileName);

            Scanner fs = new Scanner(file);

            String content = "";

            while (fs.hasNextLine()) {

                content = content + fs.nextLine() + "\n";

            }

            fs.close();

            String reverse = "";

            for (int i = content.length() - 1; i >= 0; i--) {

                char ch = content.charAt(i);

                if (Character.isUpperCase(ch))

                    reverse = reverse + Character.toLowerCase(ch);

                else if (Character.isLowerCase(ch))

                    reverse = reverse + Character.toUpperCase(ch);

                else
```

```

        reverse = reverse + ch;
    }

    System.out.println("\nFile contents in reverse order with changed case:");

    System.out.println(reverse);
} catch (FileNotFoundException e) {

    System.out.println("File not found.");

}

}

}

```

Slip 6

1) Write a program to define a class Account having members custname, accno. Define default and parameterized constructor. Create a subclass called SavingAccount with members savingbal, minbal. Create a derived class AccountDetail that extends the class SavingAccount with members, depositamt and withdrawalamt. Write a appropriate method to display customer details.

```

import java.util.Scanner;

class Account {

    String custname;

    int accno;

    Account() {

        custname = "";

        accno = 0;
    }
}

```

```
}  
Account(String custname, int accno) {  
    this.custname = custname;  
    this.accno = accno;  
}  
}  
  
class SavingAccount extends Account {  
    double savingbal;  
    double minbal;  
  
    SavingAccount(String custname, int accno, double savingbal, double minbal) {  
        super(custname, accno);  
        this.savingbal = savingbal;  
        this.minbal = minbal;  
    }  
}  
  
class AccountDetail extends SavingAccount {  
    double depositamt;  
    double withdrawamt;  
  
    AccountDetail(String custname, int accno, double savingbal,  
        double minbal, double depositamt, double withdrawamt) {  
        super(custname, accno, savingbal, minbal);  
    }  
}
```

```

        this.depositamt = depositamt;

        this.withdrawalamt = withdrawalamt;
    }

    void display() {
        double balance = savingbal + depositamt - withdrawalamt;

        System.out.println("\n--- Customer Account Details ---");

        System.out.println("Customer Name    : " + custname);
        System.out.println("Account Number  : " + accno);
        System.out.println("Saving Balance  : " + savingbal);
        System.out.println("Minimum Balance : " + minbal);
        System.out.println("Deposit Amount  : " + depositamt);
        System.out.println("Withdrawal Amount: " + withdrawalamt);
        System.out.println("Current Balance : " + balance);
    }
}

public class AccountDemo {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Customer Name: ");

        String name = sc.nextLine();

        System.out.print("Enter Account Number: ");

        int accno = sc.nextInt();

        System.out.print("Enter Saving Balance: ");

```

```

double savingbal = sc.nextDouble();

System.out.print("Enter Minimum Balance: ");

double minbal = sc.nextDouble();

System.out.print("Enter Deposit Amount: ");

double depositamt = sc.nextDouble();


System.out.print("Enter Withdrawal Amount: ");

double withdrawamt = sc.nextDouble();


AccountDetail a = new AccountDetail(
    name, accno, savingbal, minbal, depositamt, withdrawamt
);
a.display();
}
}

```

Slip 7

1)Write a class Driver with attributes license_no, name, address and age. Initialize values through the parameterized constructor. If age of Driver is less than 18 then user-defined exception should be generated —Age is below 18 years—

```
import java.util.Scanner;
```

```
class AgeException extends Exception {  
    AgeException(String message) {  
        super(message);  
    }  
}
```

```
class Driver {  
    String license_no;  
    String name;  
    String address;  
    int age;  
  
    Driver(String license_no, String name, String address, int age)  
        throws AgeException {  
  
        if (age < 18) {  
            throw new AgeException("Age is below 18 years");  
        }  
  
        this.license_no = license_no;  
        this.name = name;  
        this.address = address;  
        this.age = age;  
    }  
}
```

```
}
```

```
void display() {
```

```
    System.out.println("\n--- Driver Details ---");
```

```
    System.out.println("License No : " + license_no);
```

```
    System.out.println("Name      : " + name);
```

```
    System.out.println("Address   : " + address);
```

```
    System.out.println("Age       : " + age);
```

```
}
```

```
}
```

```
public class DriverDemo {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter License Number: ");
```

```
        String license = sc.nextLine();
```

```
        System.out.print("Enter Name: ");
```

```
        String name = sc.nextLine();
```

```
        System.out.print("Enter Address: ");
```

```
        String address = sc.nextLine();
```

```
System.out.print("Enter Age: ");

int age = sc.nextInt();

try {

    Driver d = new Driver(license, name, address, age);

    d.display();

} catch (AgeException e) {

    System.out.println("Exception: " + e.getMessage());

}

}

}
```

Slip 8

1)Write a program to create an abstract class named Shape that contains two integers and an empty method named printArea(). Provide three classes named Rectangle, Triangle and Circle such that each one of the classes extends the class Shape. Each one of the classes contain only the method printArea() that prints the area of the given shape. (use method overriding).

```
import java.util.Scanner;
```

```
abstract class Shape {

    int a, b;
```

```
    abstract void printArea();  
}
```

```
class Rectangle extends Shape {  
    Rectangle(int length, int breadth) {  
        a = length;  
        b = breadth;  
    }  
}
```

```
    void printArea() {  
        System.out.println("Area of Rectangle = " + (a * b));  
    }  
}
```

```
class Triangle extends Shape {  
    Triangle(int base, int height) {  
        a = base;  
        b = height;  
    }  
}
```

```
    void printArea() {  
        System.out.println("Area of Triangle = " + (0.5 * a * b));  
    }  
}
```

```
}
```

```
class Circle extends Shape {
```

```
    Circle(int radius) {
```

```
        a = radius;
```

```
    }
```

```
    void printArea() {
```

```
        System.out.println("Area of Circle = " + (Math.PI * a * a));
```

```
    }
```

```
}
```

```
public class ShapeDemo {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter length of rectangle: ");
```

```
        int length = sc.nextInt();
```

```
        System.out.print("Enter breadth of rectangle: ");
```

```
        int breadth = sc.nextInt();
```

```
        Rectangle r = new Rectangle(length, breadth);
```

```
r.printArea();

System.out.print("Enter base of triangle: ");
int base = sc.nextInt();

System.out.print("Enter height of triangle: ");
int height = sc.nextInt();

Triangle t = new Triangle(base, height);
t.printArea();

System.out.print("Enter radius of circle: ");
int radius = sc.nextInt();

Circle c = new Circle(radius);
c.printArea();
}
}
```

Slip 9

1)Write a program which define class Employee with data member as id, name and salary Store the information of ‘n’ employees and display the name of employee having maximum salary (Use array of object)

```
import java.util.Scanner;
```

```
class Employee {  
    int id;  
  
    String name;  
  
    double salary;  
  
    Employee(int id, String name, double salary) {  
        this.id = id;  
  
        this.name = name;  
  
        this.salary = salary;  
    }  
}
```

```
public class EmployeeDemo {  
    public static void main(String[] args) {  
  
        Scanner sc = new Scanner(System.in);  
  
        System.out.print("Enter number of employees: ");  
  
        int n = sc.nextInt();  
  
        Employee[] emp = new Employee[n];  
  
        for (int i = 0; i < n; i++) {  
  
            System.out.println("\nEnter details of Employee " + (i + 1));  
  
            System.out.print("Enter ID: ");  
  
            int id = sc.nextInt();  
  
            sc.nextLine();  
  
            System.out.print("Enter Name: ");
```

```

        String name = sc.nextLine();

        System.out.print("Enter Salary: ");

        double salary = sc.nextDouble();

        emp[i] = new Employee(id, name, salary);
    }

    int maxIndex = 0;

    for (int i = 1; i < n; i++) {

        if (emp[i].salary > emp[maxIndex].salary) {

            maxIndex = i;

        }

    }

    System.out.println("\nEmployee with Maximum Salary:");

    System.out.println("Name   : " + emp[maxIndex].name);

    System.out.println("Salary : " + emp[maxIndex].salary);

}
}

```

Slip 10

Write a program to read the contents of “abc.txt” file. Display the contents of file in uppercase as output.

Sample abc.txt

Hello World

Java Programming

Welcome to Java

Main program

```
import java.io.File;

import java.io.FileNotFoundException;

import java.util.Scanner;


public class FileUppercase {

    public static void main(String[] args) {

        try {

            File file = new File("abc.txt");

            Scanner sc = new Scanner(file);


            System.out.println("File contents in uppercase:");


            while (sc.hasNextLine()) {

                String line = sc.nextLine();

                System.out.println(line.toUpperCase());

            }


            sc.close();

        } catch (FileNotFoundException e) {

            System.out.println("File not found.");

        }

    }

}
```

```
    }  
}  
}
```

Slip 11

1)Write a program to create a super class Vehicle having members Company and price. Derive two different classes LightMotorVehicle (mileage) and HeavyMotorVehicle (capacity_in_tons). Accept the information for "n" vehicles and display the information in appropriate form. While taking data, ask user about the type of vehicle first

```
import java.util.Scanner;  
  
class Vehicle {  
    String company;  
    double price;  
    Vehicle(String company, double price) {  
        this.company = company;  
        this.price = price;  
    }  
}  
  
class LightMotorVehicle extends Vehicle {  
    double mileage;  
    LightMotorVehicle(String company, double price, double mileage) {
```

```
        super(company, price);
        this.mileage = mileage;
    }

    void display() {
        System.out.println("Vehicle Type : Light Motor Vehicle");
        System.out.println("Company    : " + company);
        System.out.println("Price      : " + price);
        System.out.println("Mileage    : " + mileage + " km/l");
    }
}

class HeavyMotorVehicle extends Vehicle {
    double capacity_in_tons;

    HeavyMotorVehicle(String company, double price, double capacity_in_tons) {
        super(company, price);
        this.capacity_in_tons = capacity_in_tons;
    }

    void display() {
        System.out.println("Vehicle Type : Heavy Motor Vehicle");
        System.out.println("Company    : " + company);
        System.out.println("Price      : " + price);
        System.out.println("Capacity    : " + capacity_in_tons + " tons");
    }
}
```

```
public class VehicleDemo {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter number of vehicles: ");  
        int n = sc.nextInt();  
        Vehicle[] vehicles = new Vehicle[n];  
        for (int i = 0; i < n; i++) {  
            System.out.println("\nVehicle " + (i + 1));  
            System.out.println("1. Light Motor Vehicle");  
            System.out.println("2. Heavy Motor Vehicle");  
            System.out.print("Enter vehicle type: ");  
            int type = sc.nextInt();  
            sc.nextLine();  
            System.out.print("Enter Company: ");  
            String company = sc.nextLine();  
            System.out.print("Enter Price: ");  
            double price = sc.nextDouble();  
            if (type == 1) {  
                System.out.print("Enter Mileage: ");  
                double mileage = sc.nextDouble();  
                vehicles[i] = new LightMotorVehicle(company, price, mileage);  
            }  
            else if (type == 2) {
```

```
        System.out.print("Enter Capacity in Tons: ");
        double capacity = sc.nextDouble();
        vehicles[i] = new HeavyMotorVehicle(company, price, capacity);
    }
    else {
        System.out.println("Invalid vehicle type.");
        i--;
    }
}
System.out.println("\n--- Vehicle Details ---");
for (int i = 0; i < n; i++) {
    if (vehicles[i] instanceof LightMotorVehicle) {
        ((LightMotorVehicle) vehicles[i]).display();
    }
    else if (vehicles[i] instanceof HeavyMotorVehicle) {
        ((HeavyMotorVehicle) vehicles[i]).display();
    }
    System.out.println();
}
sc.close();
}
```

Slip 12

1) Write a package game which will have 2 classes Indoor & Outdoor. Use a function display () to generate the list of players for the specific game. Use default & parameterized constructor

Step 1: mkdir -p JavaProject/game;

Step 2: cd JavaProject

Step 3:vim game/Indoor.java

```
package game;

public class Indoor {

    String game;

    String[] players;

    public Indoor() {

        game = "Carrom";

        players = new String[]{"Rahul", "Amit", "Priya", "Sneha"};

    }

    public Indoor(String game, String[] players) {

        this.game = game;

        this.players = players;

    }

    public void display() {

        System.out.println("Indoor Game: " + game);

        System.out.println("Players:");

        for (String p : players) {
```

```
        System.out.println(p);
    }
}
}
```

Step 4:vim game/Outdoor.java

```
package game;

public class Outdoor {

    String game;

    String[] players;

    public Outdoor() {

        game = "Cricket";

        players = new String[]{"Virat", "Rohit", "Rahul", "Hardik"};

    }

    public Outdoor(String game, String[] players) {

        this.game = game;

        this.players = players;

    }

    public void display() {

        System.out.println("Outdoor Game: " + game);

        System.out.println("Players:");

        for (String p : players) {

            System.out.println(p);

        }

    }

}
```

```
    }  
  }  
}
```

Step 5: vim GameDemo.java

```
import game.Indoor;  
  
import game.Outdoor;  
  
public class GameDemo {  
    public static void main(String[] args) {  
        Indoor i = new Indoor();  
        i.display();  
  
        System.out.println();  
  
        String[] players = {"Amit", "Sagar", "Rohit", "Akash"};  
  
        Outdoor o = new Outdoor("Football", players);  
        o.display();  
    }  
}
```

Step 6: javac game/Indoor.java game/Outdoor.java GameDemo.java

OR

javac gameDemo.java

Step 7: java gameDemo

Slip 13

1) Define a “Clock” class that does the following;

a. Accept Hours, Minutes and Seconds

b. Check the validity of numbers

c. Set the time to AM/PM mode

Use the necessary constructors and methods to do the above task.

```
import java.util.Scanner;
```

```
class Clock {
```

```
    int hours;
```

```
    int minutes;
```

```
    int seconds;
```

```
    Clock() {
```

```
        hours = 0;
```

```
        minutes = 0;
```

```
        seconds = 0;
```

```
    }
```

```
    Clock(int hours, int minutes, int seconds) {
```

```
        this.hours = hours;
```

```
        this.minutes = minutes;
```

```
        this.seconds = seconds;
```

```
    }
```

```
    boolean isValid() {
```

```
        if (hours < 0 || hours > 23)
```

```
            return false;
```

```

    if (minutes < 0 || minutes > 59)
        return false;
    if (seconds < 0 || seconds > 59)
        return false;
    return true;
}

void displayAMPM() {
    int h = hours;
    String period;
    if (h >= 12)
        period = "PM";
    else
        period = "AM";
    if (h == 0)
        h = 12;
    else if (h > 12)
        h = h - 12;
    System.out.printf("Time in AM/PM format: %02d:%02d:%02d %s",
        h, minutes, seconds, period);
}
}

```

```
public class ClockDemo {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter Hours: ");  
        int h = sc.nextInt();  
        System.out.print("Enter Minutes: ");  
        int m = sc.nextInt();  
        System.out.print("Enter Seconds: ");  
        int s = sc.nextInt();  
        Clock c = new Clock(h, m, s);  
        if (c.isValid()) {  
            System.out.println("Time is valid.");  
            c.displayAMPM();  
        } else {  
            System.out.println("Invalid time.");  
        }  
        sc.close();  
    }  
}
```

Slip 14

1) Write a JavaFX program that takes a numeric input from a user via a TextField and displays Whether it is a prime number in an output field upon clicking a “Process” button.

```
import javafx.application.Application;
```

```
import javafx.scene.Scene;
```

```
import javafx.scene.control.*;
```

```
import javafx.scene.layout.*;
```

```
import javafx.stage.Stage;
```

```
public class PrimeNumber extends Application {
```

```
    public void start(Stage stage) {
```

```
        Label inputLabel = new Label("Enter Number:");
```

```
        TextField inputField = new TextField();
```

```
        Label outputLabel = new Label("Result:");
```

```
        TextField outputField = new TextField();
```

```
        outputField.setEditable(false);
```

```
        Button processButton = new Button("Process");
```

```

processButton.setOnAction(e -> {

    int n = Integer.parseInt(inputField.getText());

    boolean prime = true;

    if (n <= 1) {

        prime = false;

    } else {

        for (int i = 2; i <= n / 2; i++) {

            if (n % i == 0) {

                prime = false;

                break;

            }

        }

    }

    if (prime)

        outputField.setText(n + " is a Prime Number");

    else

        outputField.setText(n + " is Not a Prime Number");

});

VBox root = new VBox(10);

root.getChildren().addAll(

```

```

        inputLabel, inputField,
        processButton,
        outputLabel, outputField
    );

    Scene scene = new Scene(root, 350, 250);

    stage.setTitle("Prime Number Checker");
    stage.setScene(scene);
    stage.show();
}

public static void main(String[] args) {
    launch(args);
}
}

```

Slip 15

1) Write a JavaFX program to build a simple calculator with digit buttons (0-9) and basic arithmetic operator buttons (+, -, *, /). Implement action listeners for each button (digits and operators) to perform calculations.

```

import javafx.application.Application;
import javafx.scene.Scene;

```

```
import javafx.scene.control.Button;
import javafx.scene.control.TextField;
import javafx.scene.layout.GridPane;
import javafx.stage.Stage;

public class Calculator extends Application {

    TextField display = new TextField();

    double num1 = 0;

    double num2 = 0;

    String operator = "";

    public void start(Stage stage) {

        display.setEditable(false);

        GridPane grid = new GridPane();
        grid.setHgap(5);
        grid.setVgap(5);

        Button b0 = new Button("0");
        Button b1 = new Button("1");
        Button b2 = new Button("2");
```

```
Button b3 = new Button("3");
```

```
Button b4 = new Button("4");
```

```
Button b5 = new Button("5");
```

```
Button b6 = new Button("6");
```

```
Button b7 = new Button("7");
```

```
Button b8 = new Button("8");
```

```
Button b9 = new Button("9");
```

```
Button add = new Button("+");
```

```
Button sub = new Button("-");
```

```
Button mul = new Button("*");
```

```
Button div = new Button("/");
```

```
Button equal = new Button("=");
```

```
Button[] digits = {b0, b1, b2, b3, b4, b5, b6, b7, b8, b9};
```

```
for (Button b : digits) {
```

```
    b.setOnAction(e -> {
```

```
        display.appendText(b.getText());
```

```
    });
```

```
}
```

```
add.setOnAction(e -> setOperator("+"));
```

```
sub.setOnAction(e -> setOperator("-"));
```

```
mul.setOnAction(e -> setOperator("*"));
```

```
div.setOnAction(e -> setOperator("/"));
```

```
equal.setOnAction(e -> calculate());
```

```
grid.add(display, 0, 0, 4, 1);
```

```
grid.add(b7, 0, 1);
```

```
grid.add(b8, 1, 1);
```

```
grid.add(b9, 2, 1);
```

```
grid.add(div, 3, 1);
```

```
grid.add(b4, 0, 2);
```

```
grid.add(b5, 1, 2);
```

```
grid.add(b6, 2, 2);
```

```
grid.add(mul, 3, 2);
```

```
grid.add(b1, 0, 3);
```

```
grid.add(b2, 1, 3);
```

```
grid.add(b3, 2, 3);
```

```
grid.add(sub, 3, 3);
```

```
grid.add(b0, 0, 4);  
grid.add(equal, 1, 4);  
grid.add(add, 3, 4);
```

```
Scene scene = new Scene(grid, 300, 250);
```

```
stage.setTitle("Simple Calculator");  
stage.setScene(scene);  
stage.show();  
}
```

```
void setOperator(String op) {  
    num1 = Double.parseDouble(display.getText());  
    operator = op;  
    display.clear();  
}
```

```
void calculate() {  
    num2 = Double.parseDouble(display.getText());  
  
    double result = 0;  
  
    if (operator.equals("+"))
```

```

        result = num1 + num2;
    else if (operator.equals("-"))
        result = num1 - num2;
    else if (operator.equals("*"))
        result = num1 * num2;
    else if (operator.equals("/")) {
        if (num2 == 0) {
            display.setText("Cannot divide by zero");
            return;
        }
        result = num1 / num2;
    }

    display.setText(String.valueOf(result));
}

public static void main(String[] args) {
    launch(args);
}
}

```

Slip 16

1) Write a JavaFX application that simulates a shopping cart. Add items with buttons, and implement action listeners to update the cart when an item is added

```
import javafx.application.Application;
```

```
import javafx.scene.Scene;
```

```
import javafx.scene.control.*;
```

```
import javafx.scene.layout.*;
```

```
import javafx.stage.Stage;
```

```
public class ShoppingCart extends Application {
```

```
    TextArea cartArea = new TextArea();
```

```
    Label totalLabel = new Label("Total: ₹0.00");
```

```
    double total = 0;
```

```
    public void start(Stage stage) {
```

```
        Button apple = new Button("Apple - ₹50");
```

```
        Button milk = new Button("Milk - ₹30");
```

```
        Button bread = new Button("Bread - ₹40");
```

```
        Button rice = new Button("Rice - ₹60");
```

```
        cartArea.setEditable(false);
```

```
        apple.setOnAction(e -> addItem("Apple", 50));
```

```
milk.setOnAction(e -> addItem("Milk", 30));  
bread.setOnAction(e -> addItem("Bread", 40));  
rice.setOnAction(e -> addItem("Rice", 60));
```

```
VBox items = new VBox(10);  
items.getChildren().addAll(  
    new Label("Available Items"),  
    apple, milk, bread, rice  
);
```

```
VBox cart = new VBox(10);  
cart.getChildren().addAll(  
    new Label("Shopping Cart"),  
    cartArea,  
    totalLabel  
);
```

```
HBox root = new HBox(30);  
root.getChildren().addAll(items, cart);
```

```
Scene scene = new Scene(root, 500, 350);
```

```
stage.setTitle("Shopping Cart");
```

```

        stage.setScene(scene);

        stage.show();
    }

    void addItem(String item, double price) {

        cartArea.appendText(item + " - ₹" + price + "\n");

        total = total + price;

        totalLabel.setText("Total: ₹" + total);

    }

    public static void main(String[] args) {

        launch(args);

    }
}

```

Slip 18

1)Creating an application with multiple buttons (e.g., Red, Green, Blue) that print their respective color or change the background color when clicked.

```

import javafx.application.Application;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.layout.HBox;
import javafx.stage.Stage;

public class ColorButtons extends Application {

    public void start(Stage stage) {

        Button red = new Button("Red");

```

```

Button green = new Button("Green");
Button blue = new Button("Blue");

HBox root = new HBox(20);
root.getChildren().addAll(red, green, blue);

Scene scene = new Scene(root, 400, 200);

red.setOnAction(e -> {
    root.setStyle("-fx-background-color: red;");
    System.out.println("Red");
});

green.setOnAction(e -> {
    root.setStyle("-fx-background-color: green;");
    System.out.println("Green");
});

blue.setOnAction(e -> {
    root.setStyle("-fx-background-color: blue;");
    System.out.println("Blue");
});

stage.setTitle("Color Buttons");
stage.setScene(scene);
stage.show();
}

public static void main(String[] args) {
    launch(args);
}
}

```

Slip 20

1) Design a screen to handle the Mouse Events such as MOUSE_MOVED and MOUSE_CLICK and display the position of the Mouse_Click in a TextField

```

import javafx.application.Application;

import javafx.scene.Scene;

import javafx.scene.control.Label;

import javafx.scene.control.TextField;

```

```
import javafx.scene.layout.VBox;

import javafx.stage.Stage;

public class MouseEvents extends Application {

    public void start(Stage stage) {

        Label label = new Label("Move or Click the Mouse");

        TextField textField = new TextField();

        textField.setEditable(false);

        VBox root = new VBox(15);

        root.getChildren().addAll(label, textField);


        Scene scene = new Scene(root, 500, 300);

        scene.setOnMouseMoved(e -> {

            label.setText("Mouse Position: X = " +

                (int)e.getX() + ", Y = " + (int)e.getY());

        });

        scene.setOnMouseClicked(e -> {

            textField.setText("Mouse Clicked at: X = " +

                (int)e.getX() + ", Y = " + (int)e.getY());

        });

        stage.setTitle("Mouse Event Handling");

        stage.setScene(scene);

        stage.show();

    }
```

```
public static void main(String[] args) {  
    launch(args);  
}  
}
```

Slip 21

1) Write a program to create parent class College (cno, cname, caddr) and derived class Department (dno, dname) from College. Write a necessary methods to display College details.

```
import java.util.Scanner;
```

```
class College {
```

```
    int cno;
```

```
    String cname;
```

```
    String caddr;
```

```
    void acceptCollege() {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter College Number: ");
```

```
        cno = sc.nextInt();
```

```
        sc.nextLine();
```

```
        System.out.print("Enter College Name: ");
```

```
        cname = sc.nextLine();
```

```
        System.out.print("Enter College Address: ");
```

```
        caddr = sc.nextLine();  
    }
```

```
void displayCollege() {  
    System.out.println("\nCollege Number: " + cno);  
    System.out.println("College Name: " + cname);  
    System.out.println("College Address: " + caddr);  
}  
}
```

```
class Department extends College {  
    int dno;  
    String dname;  
  
    void acceptDepartment() {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter Department Number: ");  
        dno = sc.nextInt();  
        sc.nextLine();  
        System.out.print("Enter Department Name: ");  
        dname = sc.nextLine();  
    }  
}
```

```
void displayDepartment() {  
    displayCollege();  
    System.out.println("Department Number: " + dno);  
    System.out.println("Department Name: " + dname);  
}  
}
```

```
public class CollegeDemo {  
    public static void main(String[] args) {  
        Department d = new Department();  
        d.acceptCollege();  
        d.acceptDepartment();  
        System.out.println("\n--- College Details ---");  
        d.displayDepartment();  
    }  
}
```